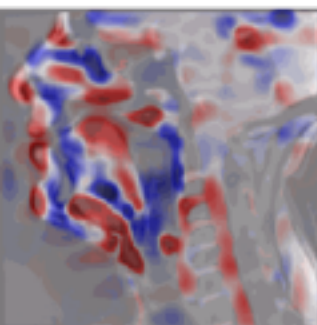
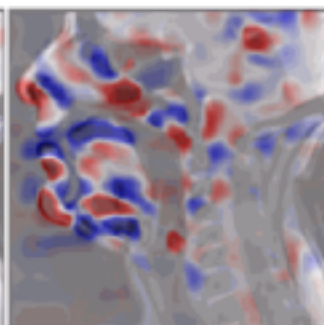


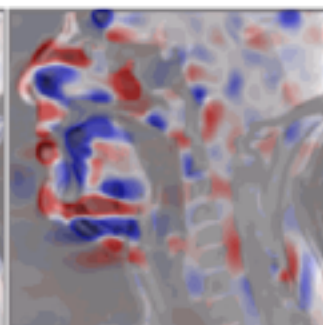
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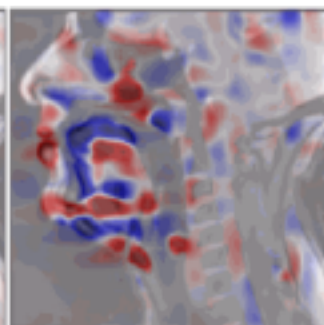
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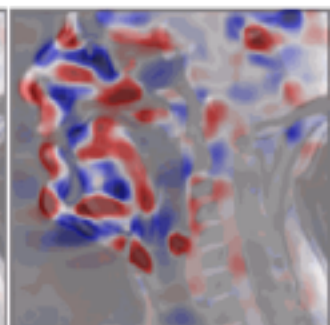
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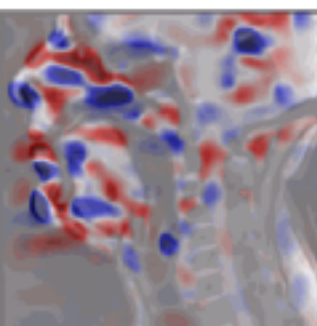
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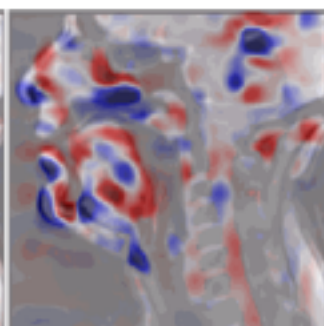
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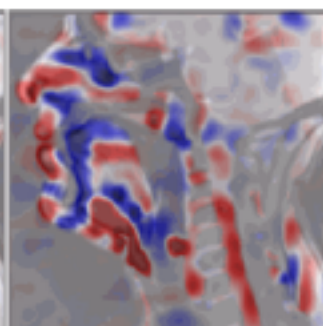
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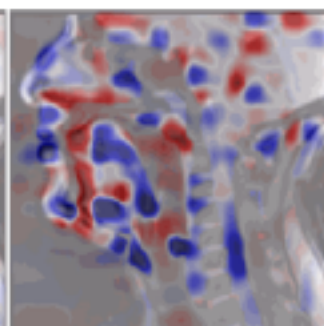
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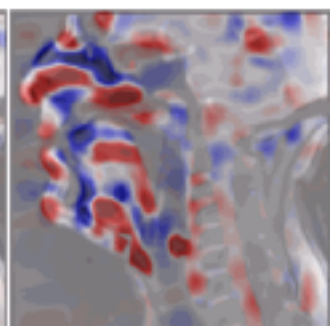
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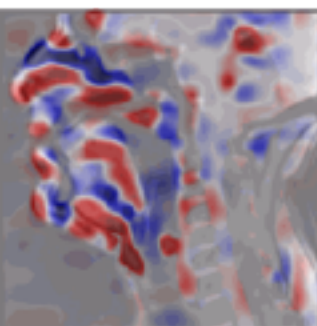
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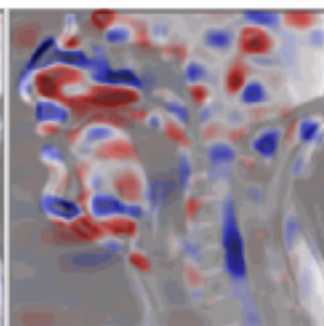
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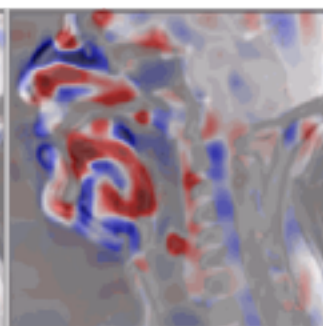
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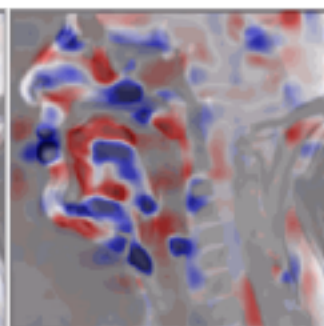
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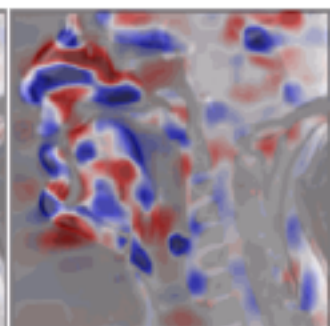
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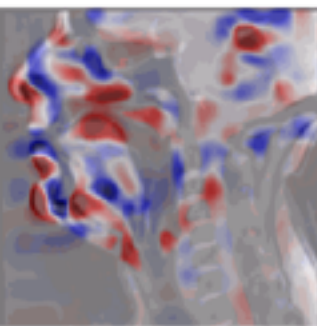
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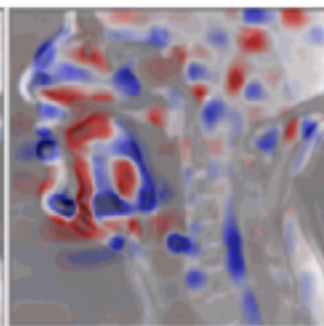
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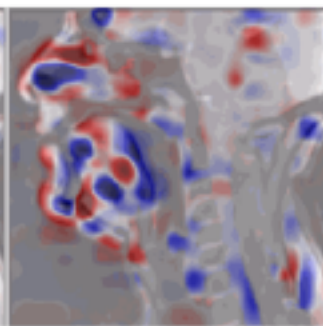
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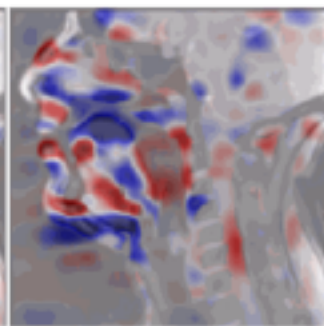
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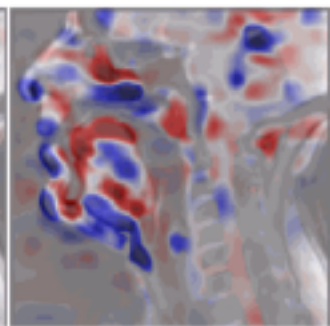
18



19



20



Methods

Distinctives

1

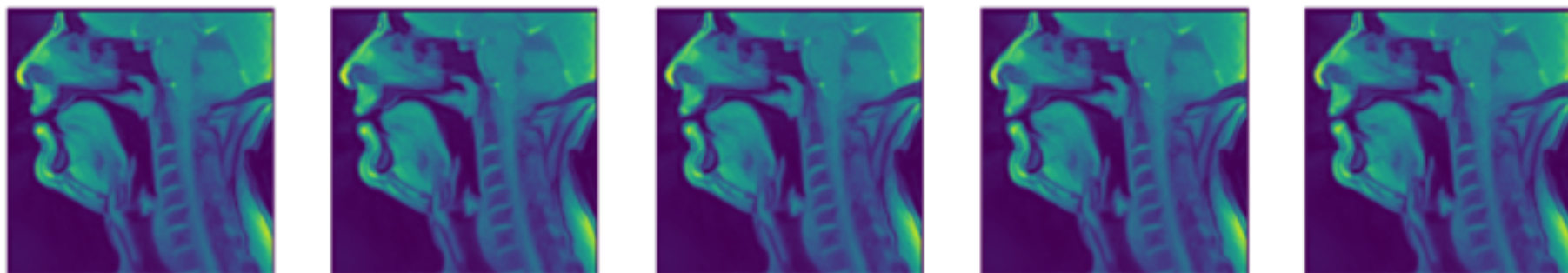
8

• Extracted featur matrix of /l/ in *filou* /filu/ “trickster”
produced by participant P6 in FR-MRI.

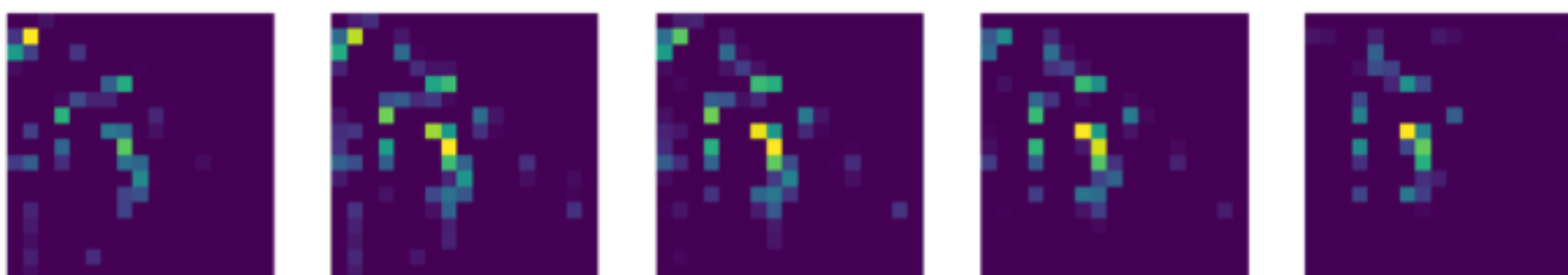
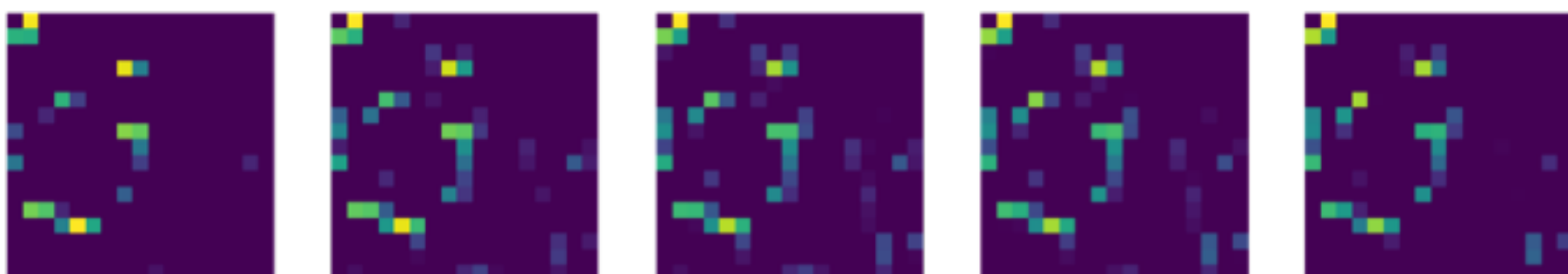
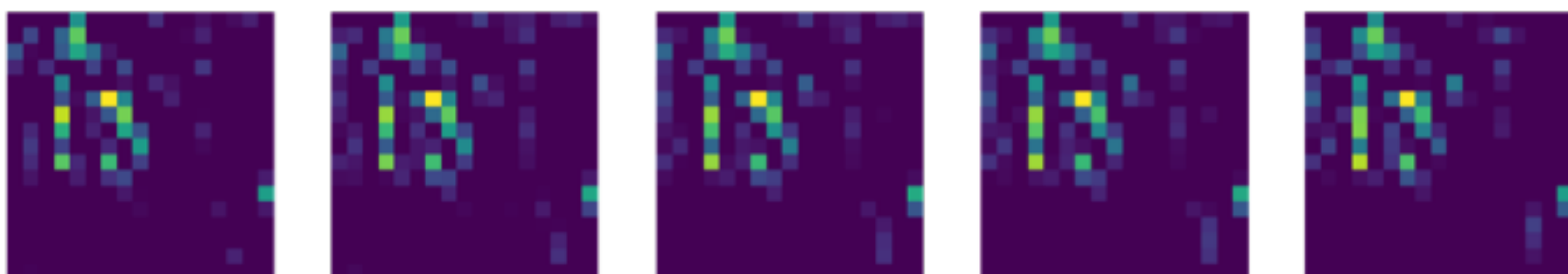
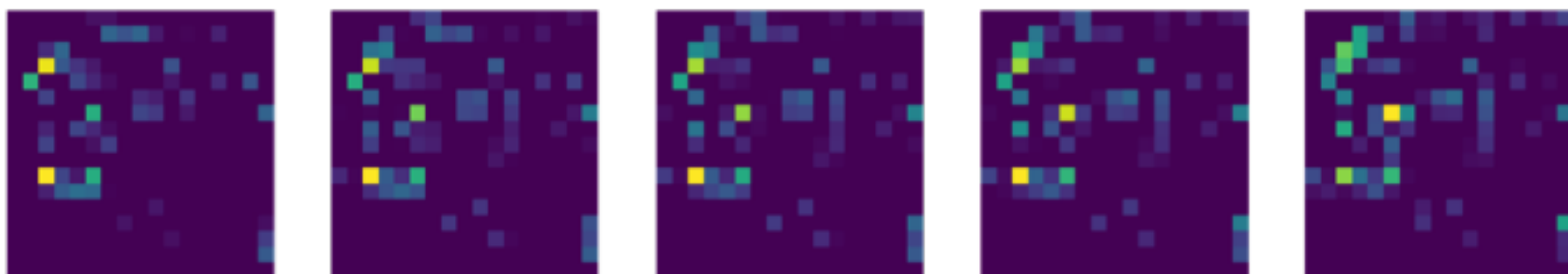


USOC

Time



Encoder
(3D CNN)



Feature

Tine







M

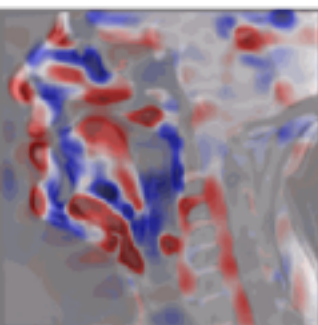
L

P

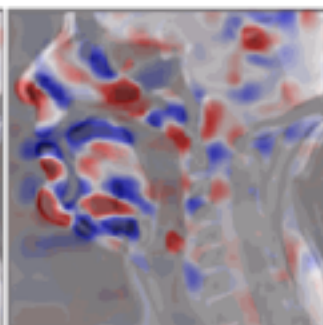
Latent dimensions (shown as saliency maps)



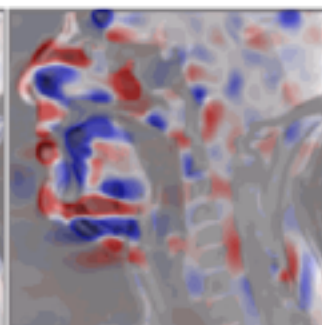
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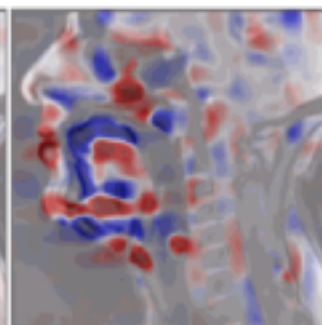
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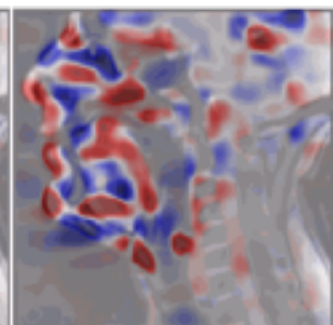
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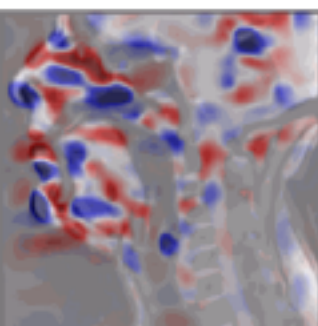
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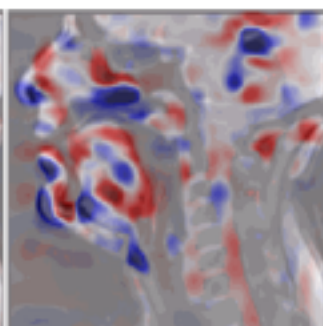
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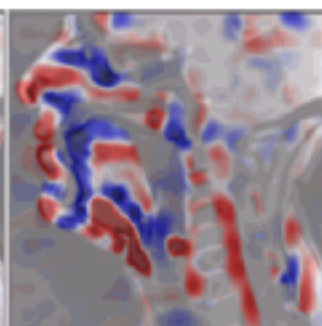
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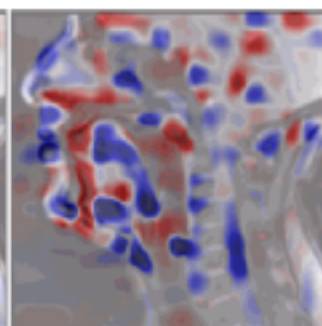
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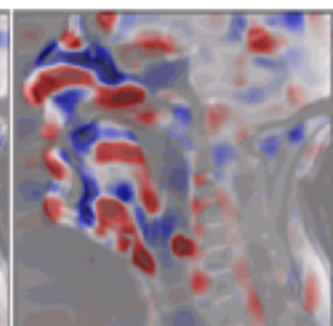
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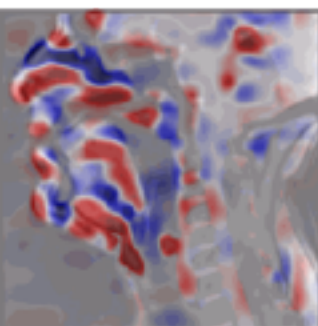
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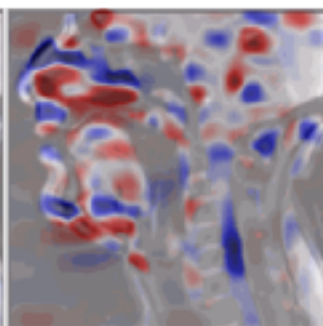
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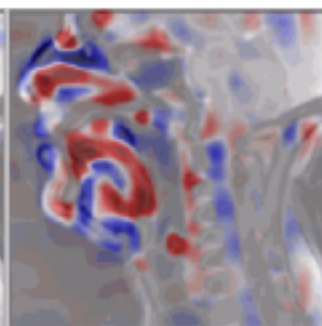
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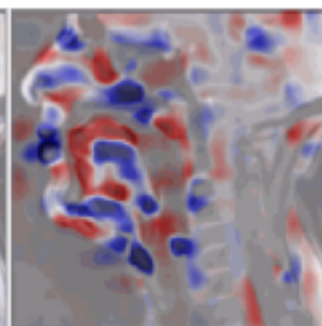
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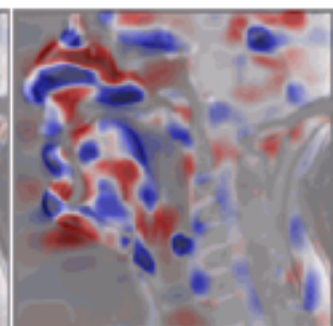
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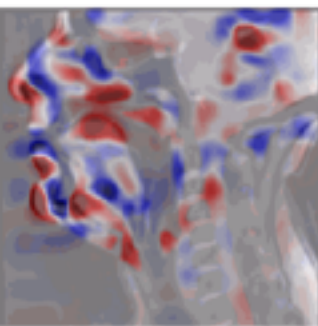
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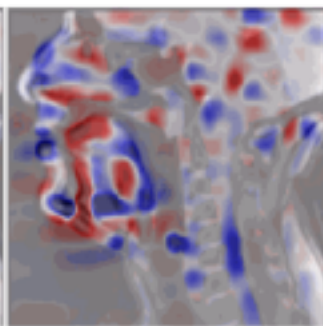
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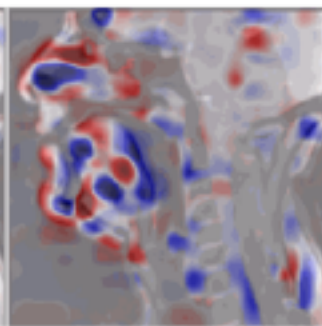
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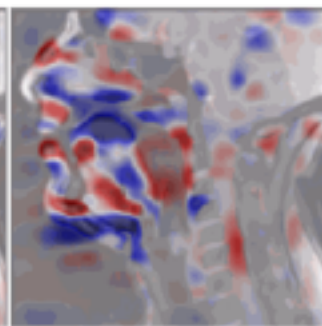
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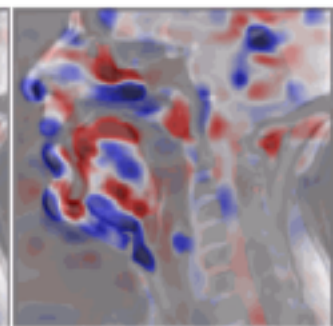
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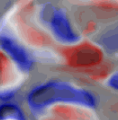


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20





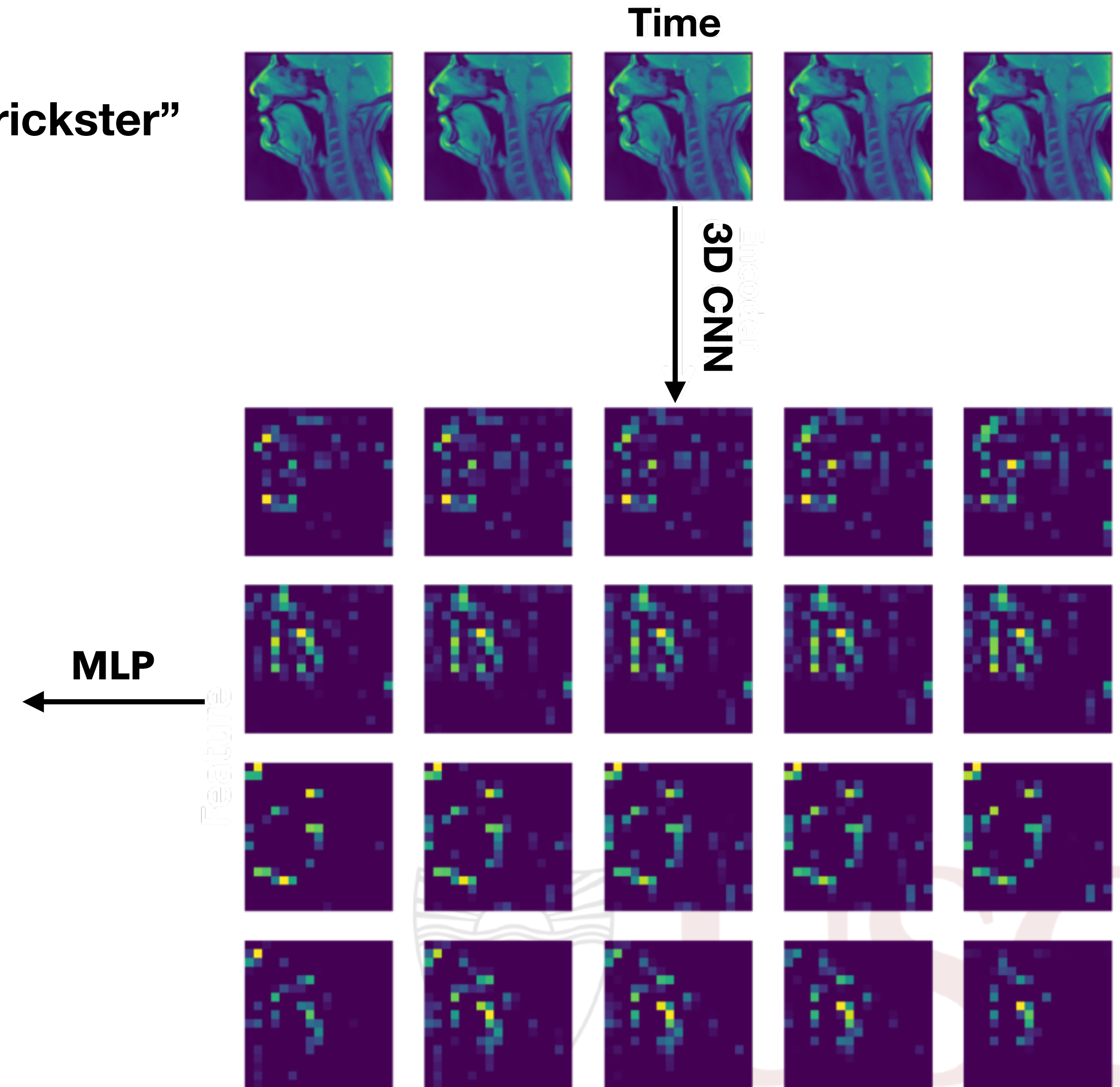
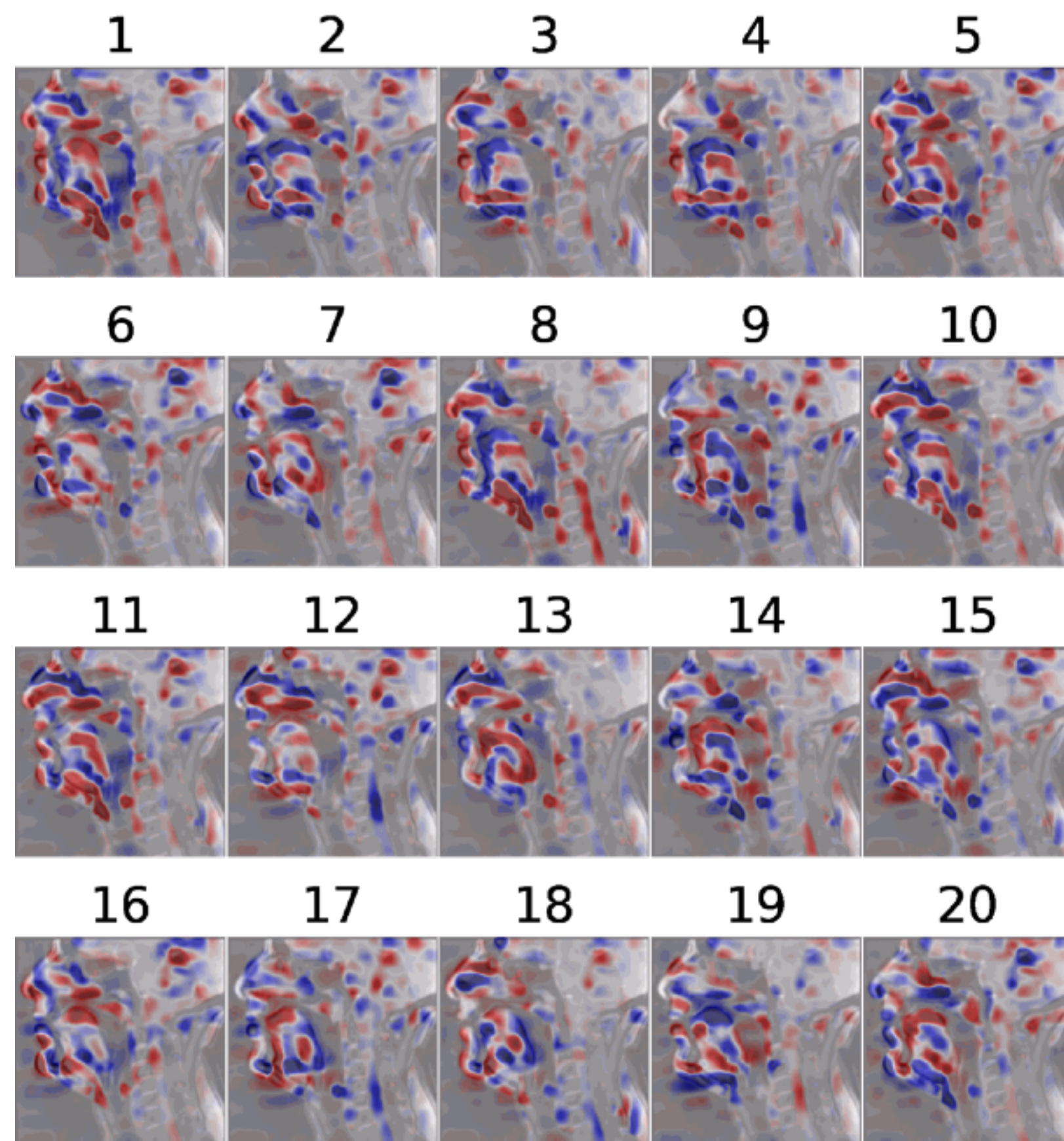


Methods

Distinctiveness

- Extracted feature matrix of /l/ in *filou* /filu/ “trickster” produced by participant P6 in FR-MRI.

Latent dimensions (shown as saliency maps)





Methods

Preview

1. Data: acoustic/articulatory data from published datasets
2. Distinctiveness measurement in acoustics & articulation
- 3. Bigram surprisal calculation**
4. Phoneme flexibility: degree of freedom between articulatory synergies & the speech target



USC